

Research Plan — Minimax Estimation on Orbit Spaces: the Exact Rate Gain from Invariance via Orbital MDL and Concentration in the Multiplicative Twin Space

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(plan / extended abstract — not for submission as is)

June 19, 2026

Abstract

This is a *plan*, not a finished article: it fixes the target theorems and the proof architecture, reusing the building blocks already validated and corrected (orbital Glivenko–Cantelli/Donsker; Barron–Cover/Hellinger oracle inequality; Fano lower bound). The thesis is that orbit-based inference does carry genuine added value, but *not* where it was originally placed. A global gauge conjugation is an isomorphism and cannot improve a convergence theorem; the value appears only when the orbit structure genuinely reduces the *effective dimension* of the problem. We make this precise: for a G -invariant target on E , the minimax risk is governed by $d_{\text{eff}} = \dim(E/G) = \dim E - \dim(\text{orbit})$, and is achieved adaptively by an *orbital minimum-description-length* estimator. The three tools of the program are slotted in their correct roles: (1) data complexity $\rightarrow$ *metric entropy of the quotient*; (2) candidate complexity $\rightarrow$ *orbital approximation / description length*; (3) concentration in twin spaces $\rightarrow$ the *multiplicative* (logarithmic-twin) space of Hellinger affinities, where the Barron–Cover partition-function bound is a genuine large-deviation estimate. The *distinctive contribution* we target is the combination of the rate gain with a *multiplicative/heavy-tailed* data regime (positive intensities, compositional data): there the affinity coordinate is canonical and scale-invariant, and the orbital MDL estimator attains the same gained rate where identity-coordinate concentration fails — a regime outside the existing invariance-gain literature. A frank novelty assessment versus the recent “sample-complexity gain from invariance” literature closes the plan.

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1 The conceptual correction (why the program is sound but was misplaced)

A twin space $(E, \odot_\phi)$ is the image of a classical structure under a bijection ϕ (the Transfer Theorem). Hence any concentration/large-deviation statement “in the twin space” is the exact push-forward of a classical one: conjugation by a bijection is an isomorphism and creates no information, tightens no bound, and cannot lower the effective sample size below the number of independent observations. This single fact explains every error corrected in the source notes (the spurious $\sqrt{m}$ sample inflation; the unconditional KL oracle inequality; the sup orbital metric).

Where the value actually is. Two places, both used below:

- (i) *Dimension reduction.* When G acts with *positive-dimensional* orbits, the invariant problem genuinely lives on the lower-dimensional quotient E/G . The minimax rate then improves from $n^{-2s/(2s+D)}$ to $n^{-2s/(2s+d_{\text{eff}})}$. For a *finite* group the orbits are 0-dimensional, $d_{\text{eff}} = D$, and the gain is only a constant factor — exactly consistent with the corrected orbit-averaging theorem (variance reduced by at most $|G|$, never the rate).
- (ii) *The right twin space for concentration.* The Hellinger affinity $\rho(P, Q) = \int \sqrt{dP dQ}$ is *multiplicative* over products, $\rho(P^{\otimes n}, Q^{\otimes n}) = \rho(P, Q)^n$. Thus $-\log \rho$ is additive: the natural “twin sum” of evidence is the multiplication of likelihood ratios, i.e. addition in the logarithmic-twin coordinate. The Barron–Cover/Kraft partition-function bound is precisely a Cramér-type large-deviation estimate in *this* twin space — not in the data space. This is the legitimate form of the third pillar.

2 Setting and standing assumptions

Assumption 1 (Geometry and group). (E, ρ_E) is a compact metric space (typically a compact Riemannian manifold of dimension D). A compact group G acts continuously on E by isometries, with closed orbits; generic orbits have dimension $r := \dim(G \cdot x)$. The quotient E/G is a compact metric space for the *orbital (infimum) metric*

$$d_G([x], [y]) := \inf_{g \in G} \rho_E(x, g \cdot y),$$

(the corrected metric — the supremum form is not well defined on the quotient), with covering dimension $d_{\text{eff}} := \dim(E/G) = D - r$ (for a smooth proper action).

Assumption 2 (Reference measure and invariant class). Fix a G -invariant base measure μ on E with push-forward $\mu_G := \pi_* \mu$ on E/G , $\pi : E \rightarrow E/G$. The target is *known to be G -invariant*, $f^* = \tilde{f}^* \circ \pi$, and $\tilde{f}^*$ lies in the Hölder (or Sobolev) ball $\Sigma(s, L)$ of order $s > 0$, radius L , on $(E/G, d_G, \mu_G)$. We write $\Sigma_{\text{inv}}(s, L)$ for the pulled-back invariant class on E .

Remark 2.1. Assumption 2 (the target is invariant) is the *structural prior* that makes the gain real; without it, orbit projection estimates only the invariant part $\pi_* P^*$ and there is nothing to gain.

3 Tool 1 — data complexity as metric entropy of the quotient

Lemma 3.1 (Orbital covering numbers). *Under Assumptions 1–2, the invariant class obeys*

$$\log N(\varepsilon, \Sigma_{\text{inv}}(s, L), \|\cdot\|_\infty) \asymp \left(\frac{L}{\varepsilon}\right)^{d_{\text{eff}}/s},$$

governed by $d_{\text{eff}} = \dim(E/G)$, not by D .

Proof sketch. A G -invariant function on E is a function on E/G ; cover $(E/G, d_G)$ by $\asymp (1/\varepsilon)^{d_{\text{eff}}}$ balls and apply the standard entropy bound for Hölder balls on a d_{eff} -dimensional metric space. This is the corrected covering-number lemma of the orbital-estimation note, restricted to the quotient. $\square$

4 Tool 2 — candidate complexity by orbital approximation (description length)

Lemma 4.1 (Orbital sieve and twin Jackson bound). *There is a countable family of G -invariant models $\{\mathcal{M}_m\}_{m \in \mathcal{I}}$ (invariant histograms/wavelets at resolution 2^k pulled back from E/G ; $m = (k, \theta)$) with*

$$\dim \mathcal{M}_k \asymp 2^{k d_{\text{eff}}}, \quad \inf_{q \in \mathcal{M}_k} \text{KL}(P^* \| Q) \leq C L^2 2^{-2ks} \quad (f^* \in \Sigma(s, L)),$$

and description lengths $L(m)$ satisfying Kraft $\sum_m 2^{-L(m)} \leq 1$ with $L(k, \theta) \asymp \dim \mathcal{M}_k \cdot \log(\cdot)$.

Proof sketch. Pull a standard linear sieve on E/G back through π ; invariance is automatic. The approximation (Jackson) bound is the classical one on the d_{eff} -dimensional quotient. Coding the resolution k and the cell counts gives a prefix code; replace L by $2L$ to secure the half-exponent Kraft condition $\sum 2^{-L(m)/2} \leq 1$ used in Tool 3. This is the orbital realization of “choosing the candidate function from the orbit structure.” $\square$

5 Tool 3 — concentration in the multiplicative twin space

Proposition 5.1 (Affinity is the twin sum of evidence). *For the Hellinger affinity $\rho(P, Q) = \int \sqrt{dP dQ} d\mu$, $\rho(P^{\otimes n}, Q^{\otimes n}) = \rho(P, Q)^n$; equivalently $-\log \rho(P^{\otimes n}, Q^{\otimes n}) = n(-\log \rho(P, Q))$. Concentration of log-likelihood ratios is therefore additive in the logarithmic-twin coordinate, and the Kraft partition-function bound*

$$\mathbb{E}_{P^*} \left[2^{-L(\widehat{m})/2} \prod_{i=1}^n \sqrt{\widehat{f}(X_i)/f^*(X_i)} \right] \leq \sum_m 2^{-L(m)/2} \rho(f^*, f_m)^n \leq 1$$

is a Cramér-type large-deviation estimate in the multiplicative twin space $(\mathbb{R}_{>0}, \times)$.

This is the corrected, well-posed third pillar: the relevant non-additive structure is the multiplicative space of affinities/likelihood ratios, where the “twin sum” genuinely is multiplication.

Remark 5.2 (Why the affinity, and not a free coordinate choice). The companion analysis (`concentration_diffeomo`) establishes two facts that make Proposition 5.1 the *only* legitimate form of the third pillar. First, *no free lunch*: for a fixed functional the bounded-difference/sub-Gaussian constants are *coordinate-intrinsic*, so a gauge change cannot tighten a concentration bound (and the unnormalised optimum $\inf_\phi F[\mu, \phi] = 0$ is degenerate). Hence concentration cannot, by itself, improve a convergence rate — which is exactly why the rate gain here comes from dimension reduction (Tools 1–2), not from Tool 3. Second, the affinity ρ is the canonical *scale-invariant* concentration object: $-\log \rho$ is additive and unit-free, curing the scale-dependence that invalidated coordinate “optimisation”. Tool 3 therefore serves to make the oracle inequality self-contained, not to create rate.

6 Main results (targets)

Theorem 6.1 (Upper bound: adaptive orbital MDL). *Let $\hat{f} = f_{\hat{m}}$ be the orbital MDL estimator*

$$\hat{m} \in \arg \min_{m \in \mathcal{I}} \left\{ -\sum_{i=1}^n \log f_m(X_i) + L(m) \log 2 \right\}.$$

Then, for every $f^ \in \Sigma_{\text{inv}}(s, L)$ and every $s \in (0, s_{\max}]$,*

$$\mathbb{E}[h^2(P^*, \hat{f})] \leq C \inf_k \left\{ L^2 2^{-2ks} + \frac{2^k d_{\text{eff}} \log n}{n} \right\} \leq C' \left(\frac{\log n}{n} \right)^{2s/(2s+d_{\text{eff}})},$$

adaptively (the estimator does not use s); the $\log n$ is the two-stage description-length / adaptation cost, and is removed (exact rate, adaptively) by the projection estimators of `orbital_upper_bound_proof` §6.

Proof sketch. The corrected Barron–Cover argument (Proposition 5.1) yields the Hellinger oracle inequality $\mathbb{E}[h^2(P^*, \hat{f})] \leq C \inf_m \{ \inf_{q \in \mathcal{M}_m} \text{KL}(P^* \| q) + \log 2 L(m)/n \}$. Insert the bias (Lemma 4.1) and $L(m)/n \asymp 2^{kd_{\text{eff}}}/n$, then optimise k : $2^{k^*} \asymp n^{1/(2s+d_{\text{eff}})}$. Adaptivity is automatic because the penalised criterion selects k . $\square$

Theorem 6.2 (Lower bound: Fano on the quotient).

$$\inf_{\hat{f}} \sup_{f^* \in \Sigma_{\text{inv}}(s, L)} \mathbb{E}[h^2(P^*, \hat{f})] \geq c n^{-2s/(2s+d_{\text{eff}})}.$$

Proof sketch. Build $M \asymp \exp(2^{kd_{\text{eff}}})$ invariant perturbations $f_\omega = f_0(1 + \delta \sum_j \omega_j \psi_{k,j})$ with $\omega \in \{\pm 1\}^N$, $N \asymp 2^{kd_{\text{eff}}}$, where the bumps $\psi_{k,j}$ are supported on disjoint balls of the *quotient* $(E/G, d_G)$ pulled back to E (hence automatically G -invariant, by the *infimum* metric of Assumption 1). A Varshamov–Gilbert subset gives pairwise Hellinger separation $\beta \asymp \delta^2 2^{-kd_{\text{eff}}} \cdot N \asymp \delta^2$ and pairwise $\text{KL} \leq \alpha$ with $n\alpha \asymp n\delta^2$. Using the *corrected* Fano bound (the $n\alpha$ factor, not α), balance $\delta^2 \asymp 2^{-2ks}$ and $2^k \asymp n^{1/(2s+d_{\text{eff}})}$. $\square$

Theorem 6.3 (Exact rate gain from invariance). *Under Assumptions 1–2,*

$$\boxed{\mathcal{R}_n^*(\Sigma_{\text{inv}}(s, L)) \asymp n^{-2s/(2s+d_{\text{eff}})}, \quad d_{\text{eff}} = \dim(E/G) = D - r.}$$

The invariant rate strictly improves on the non-invariant rate $n^{-2s/(2s+D)}$ iff the orbits are positive-dimensional ($r > 0$); for finite G ($r = 0$) the gain is only a constant factor.

Remark 6.4 (Exact rate, adaptively: the log is removed). $\mathcal{R}_n^*$ denotes the minimax rate, and the box is now an honest two-sided match. The lower bound (Theorem 6.2, proved in `orbital_lower_bound_proof`) is $n^{-2s/(2s+d_{\text{eff}})}$ exactly. The *adaptive* orbital MDL of Theorem 6.1 attains it up to a $\log n$ two-stage/coding cost; but this log is *not* intrinsic and is removed in `orbital_upper_bound_proof` (§6): a single-resolution *projection* estimator attains $n^{-2s/(2s+d_{\text{eff}})}$ exactly for known s , and a *penalised projection* estimator (Birgé–Massart) attains it *adaptively*. The reason is structural — a projection / penalised estimator pays the model’s metric complexity $D_k/n \asymp 2^{kd_{\text{eff}}}/n$, whereas the MDL code additionally pays $\log(1/\delta) \asymp \log n$ bits to quantise each coefficient; the geometric growth $D_k \asymp 2^{kd_{\text{eff}}}$ keeps the selection weights summable with linear-in- D_k exponents, so no log re-enters. Hence the box holds at the *exact* rate, both for known s and adaptively.

7 Examples calibrating the gain (and the honest negative cases)

- (a) **Radial densities.** $E = \mathbb{R}^D$ (or S^D), $G = SO(D)$: $d_{\text{eff}} = 1$, rate $n^{-2s/(2s+1)}$ vs. $n^{-2s/(2s+D)}$ — maximal gain.
- (b) **Partial translation invariance.** $E = \mathbb{T}^D$, $G = \mathbb{T}^k$: $d_{\text{eff}} = D - k$.
- (c) **Finite symmetry (negative control).** Compositional data on the simplex with $G = S_d$, or m -fold rotational symmetry $G = \mathbb{Z}_m$: $r = 0$, *no rate change*, only a constant $(1/|G|)$ improvement — exactly the corrected orbit-averaging statement. This example is kept deliberately, to show the theory predicts *when there is no gain*.
- (d) **Deformed-operation variant.** On positive/compositional data the natural “addition” is a twin operation $\odot$ (log-ratio, Einstein addition); $\Sigma(s, L)$ is then smoothness in the twin coordinate. The same theorems hold with the gauge built into π , illustrating the non-additive reach.

8 The multiplicative / heavy-tailed extension (the distinctive contribution)

This is the angle that sets the paper apart from the existing invariance-gain literature, which works in L^2 /Sobolev/RKHS over Euclidean or bounded domains. Here the data are *positive* or *compositional*, the natural group acts through a twin (non-additive) operation, and the identity coordinate admits no usable concentration; yet the gained rate persists.

Assumption 3 (Multiplicative regime). $E \subseteq (0, \infty)^D$ (or the simplex Δ^{D-1}), with the gauge $\phi = \log$ (resp. the centred log-ratio) so that the base operation is the twin sum $x \odot y = \phi^{-1}(\phi(x) + \phi(y))$. The group G acts by isometries of the twin coordinate $\phi(E)$, with orbital metric and quotient as in Assumption 1 computed in the ϕ -chart; $d_{\text{eff}} = \dim(\phi(E)/G)$. Smoothness $\Sigma(s, L)$ is measured in the twin coordinate.

Lemma 8.1 (Matched-coordinate concentration; from the corrected concentration note). *Let the per-coordinate (log-)ranges be $\log(b_i/a_i)$. Then for the multiplicative aggregate the deviation of log concentrates at the dimensionless scale $\sum_i \log^2(b_i/a_i)$, and for heavy-tailed positive data with sub-Gaussian $\log X$ the affinity bound of Proposition 5.1 holds in the ϕ -chart even when no identity-coordinate (Hoeffding/Bernstein) bound exists. The improvement over the identity chart is qualitative (finite vs. infinite), not a numerical factor on a common functional.*

Proof sketch. Apply Theorems “product”/“heavy” of `concentration_diffeomorphism_corrected` in the ϕ -chart; the affinity is computed with respect to the twin-transported reference measure, and Proposition 5.1 is unchanged because ρ is invariant under the change of variables. $\square$

Theorem 8.2 (Orbital MDL on multiplicative data). *Under Assumptions 1,2,3, the orbital MDL estimator built on the ϕ -chart sieve satisfies, for $f^* \in \Sigma_{\text{inv}}(s, L)$,*

$$\mathbb{E}[h^2(P^*, \hat{f})] \leq C' \left(\frac{\log n}{n} \right)^{2s/(2s+d_{\text{eff}})},$$

adaptively, with $d_{\text{eff}} = \dim(\phi(E)/G)$. The rate is identical to the additive case (Theorem 6.1); the contribution is that it holds on positive/compositional, possibly heavy-tailed, data where identity-coordinate methods give no bound — because the affinity oracle step is tail-free (`orbital_upper_bound_proof`, Lemma on the partition bound).

Proof sketch. Run the proof of Theorem 6.1 verbatim in the ϕ -chart: Lemma 3.1 (quotient entropy) and Lemma 4.1 (twin Jackson) are stated metrically and transport through ϕ ; the oracle inequality uses the scale-invariant affinity bound (Proposition 5.1, Lemma 8.1); optimise k . The lower bound (Theorem 6.2) transports identically since h^2 is invariant under the change of variables. $\square$

Remark 8.3 (What is genuinely new here). The rate gain itself (Theorem 6.3) overlaps known results; what is new is (a) the *adaptive MDL / Hellinger* route, and (b) Theorem 8.2: the gain on *multiplicative / heavy-tailed* data via the canonical affinity coordinate, a regime the additive invariance-gain literature does not cover. Concrete instances: positive Poisson/point-process intensities, compositional (simplex) data, and multiplicative-noise diffusions — the very models of the corrected MDL paper.

9 Paper outline (once the plan is executed)

1. Introduction: invariance as effective-dimension reduction; the multiplicative-twin view of evidence.
2. Geometry of the quotient and the orbital metric (Assumption 1).
3. Tool 1: quotient entropy (Lemma 3.1).
4. Tool 2: orbital sieve, twin Jackson, description length (Lemma 4.1).
5. Tool 3: affinity concentration (Proposition 5.1).
6. Upper bound, adaptivity (Theorem 6.1).
7. Lower bound (Theorem 6.2); exact rate (Theorem 6.3).
8. Examples; finite-group negative control.
9. **Multiplicative/heavy-tailed extension** (Section 8): the distinctive contribution — matched affinity coordinate, orbital MDL on positive/compositional data (Theorem 8.2).
10. Discussion: deformed operations; relation to invariant kernel methods.

10 Frank novelty assessment (read before writing)

- **What is *not* new.** The bare statement “invariance lowers the effective dimension to $\dim(E/G)$ ” is, in the L^2 /Sobolev and kernel-ridge settings, essentially established (Tahmasebi & Jegelka, *The exact sample complexity gain from invariances*, NeurIPS 2023; Bietti, Venturi, Bruna; Elesedy & Zaidi). Theorem 6.3 in those settings would be a reformulation and should not be sold as new.
- **What can be genuinely new.** (i) The *information-theoretic / MDL* route: an *adaptive* orbital description-length estimator achieving the gain in *Hellinger* risk, with the rate emerging from the Kraft partition-function bound rather than from RKHS spectra. (ii) The extension to *deformed (twin) operations*, where the invariance is with respect to a non-additive group of operations and smoothness is measured in the twin coordinate — a regime outside the cited works. (iii) The unifying *multiplicative-twin-space* formulation of concentration (Proposition 5.1) tying the program’s third pillar to a standard large-deviation principle.

- **Recommendation.** Build the paper around (i)+(ii) with Theorem 8.2 (Section 8) as the headline: *adaptive orbital MDL for invariant density estimation on multiplicative / heavy-tailed data*. Theorem 6.3 calibrates the gain (and is positioned explicitly as a re-derivation in the cited additive setting), while the multiplicative regime is the part with clear, uncontested added value. The corrected concentration note supplies the self-contained, scale-invariant Tool 3 (Remark 5.2).

Remaining work / risks

- Make Lemma 4.1 fully rigorous on a curved quotient (orbifold points where the action is not free: E/G may be a stratified space, and d_{eff} is the dimension of the top stratum).
- Confirm the Hellinger oracle constant and the half-Kraft normalisation (Tool 3) for the orbital sieve.
- Decide the deformed-operation scope (which gauges ϕ keep $\Sigma(s, L)$ a genuine smoothness class).
- Literature due diligence on the exact-gain papers to pin the precise delta.
- Multiplicative regime (Section 8): integrability of the twin-transported reference measure, and conditions on the tails of $\log X$ ensuring the affinity bound and the twin Jackson estimate both hold.